Cattle Behaviour Classifier

Members: Abdul Muiz and Muhammad Haris

https://github.com/mbyx/cattle_classifier

Task

We were tasked to train an AI model that was capable of classifying the behaviours of cattle in a field, namely, eat, drink, stand. To this end we were also required to create a user interface that served as a dashboard, showing the healthiness of cows, as well as plots of their behaviours. It would also allow registration of cattle by their ear tag as well as a way to register behaviours and tags by image or video.

Finally, we were also required to comment on the best camera angle to use for such a model for maximum accuracy.

(Note that the repository contains all code for both the model and the UI.)

Initial Research

Dataset

Our first goal was to find datasets that could be used for training the AI model. We were able to find three such datasets, Beef Cattle Behaviour Dataset [1], CVBD-5 [2], and CVB [3].

[2] and [3] came with a research paper that we read through to figure out some recommended approaches, as well as how the dataset was structured so that it could be parsed.

Using the information present in the papers, as well as the datacard in [1], we were able to determine that the best results were obtained by the Beef Cattle Behaviour Dataset, which we used as our main dataset, with the other two datasets used as an aid.

The dataset recommended the use of data augmentation and oversampling in order to counteract the data imbalance (specifically when it came to drinking data), which we implemented using random oversampling and the built-in OpenCV data augmentation.

Prior Work

We were able to acquire most of our knowledge using a literature review [4], which had a table that summarized the list of methods used to train an AI model to classify certain behaviours. We found that the most common methods were CNN/RCNN as well as YOLO. Thus we decided to prefer YOLOv26n for training our model (after comparing with YOLOv8n).

From this table we then chose two research papers for further information.

The first paper, [5], recognized that behaviour recognition can be done with both contact and contactless methods, the former requiring an edge device attached to the cattle that logs data such as accelerometer readings and either processes it locally or sends it to a common server for processing. The contactless approach makes use of low resolution video cameras that are aligned in either front angle or side angle view, and data is sent to a single server for processing. This specific paper realized

that the region of interest is not the entire cow itself but rather the mouth area of the cow as this is what shows whether the cow is eating, drinking, or nothing at all. So they took the videos, extracted frames from them and cropped with varying sizes the mouth areas of the cows, then interpolated the images to have a single fixed size.

While we did not implement this, we recognize that this could be used for future improvement.

The second paper, [6], had a large amount of useful information. Their model was trained on multi-modal data, from two cameras, one of which was top view and the other was side view. They found that the side view camera performed better than the top view camera, for every behaviour aside from drinking. It would usually be unable to distinguish between drinking and standing in side view (a problem that we also encountered). As they had data available from both views, they applied post processing to the output of the model, such that if the model trained on side view data said drinking, whereas the model trained on top view data said standing, it would refer the top view model.

As our dataset did not have both views available, we were unable to implement their post processing method. A future improvement would be to gather our own dataset from both views and combine them as explained in the paper.

This last paper thus gave us the following information:

- Side view is preferred for cow behaviour classification, however, having both views available increases accuracy. Specifically, the side view should allow a good view of the eating and drinking troughs, as well as the overall body of the cow.

- Using two models and combining their results in a post-processing step can yield greater results.

- YOLOV8 models are good enough for this classification task.

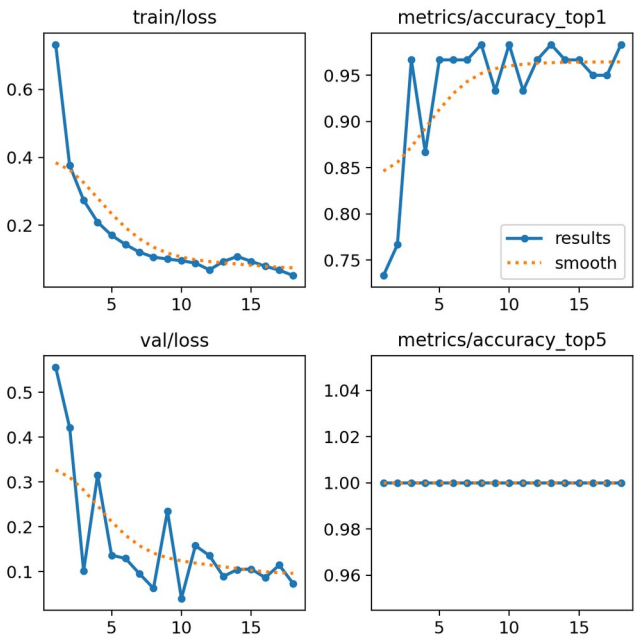
We noted that none of the papers attempted a generalized model (i.e. they did not test it on data that did not come from their own dataset). Thus if this were to be used in the real world, preferably each farm would want to have their own dataset for maximum accuracy.

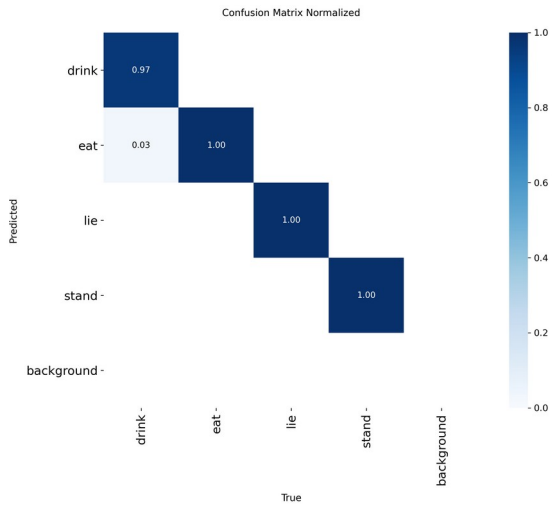
Work Details

AI Model

We made use of Kaggle's online editor for most of our work as it had a fast internet speed to easily download the large dataset, as well as GPU capabilities to speed up training. The first task was to take the dataset and split it into three categories, training data, validation data, and testing data. We used a 70:15:15 split. This data was also further grouped into three categories, eating, drinking, and nothing.

Then we began to train the AI model, but found that while it had great validation and testing accuracy, it failed to generalize very well, especially for drinking and standing. To combat this, we employed an oversampling approach as well as data augmentation. We made use of early stopping as well as tuning the hyper parameters to reduce the amount of over fitting. We ran for 50 epochs with a patience of 10. The results and confusion matrix are as follows:





User Interface

Our next task was to create the UI interface for the classifier. We chose Streamlit for this task. We make use of three models, one is the standard yolov26m model to take a video/image input and extract the bounding boxes for each of the cows, so that our classifier model that we trained can classify the behaviour, drawing the bounding box as well as the behaviour over-layed on top of the video. We also use an ear tag detection model provided by another group to extract bounding boxes of the tags from the image and apply OCR to them to extract the tag id.

We created a sort of dashboard that allows us to view the overall health of every cow as well as the real time behaviours known currently, as well as a dashboard to view individual details of each cow by searching by their tag.

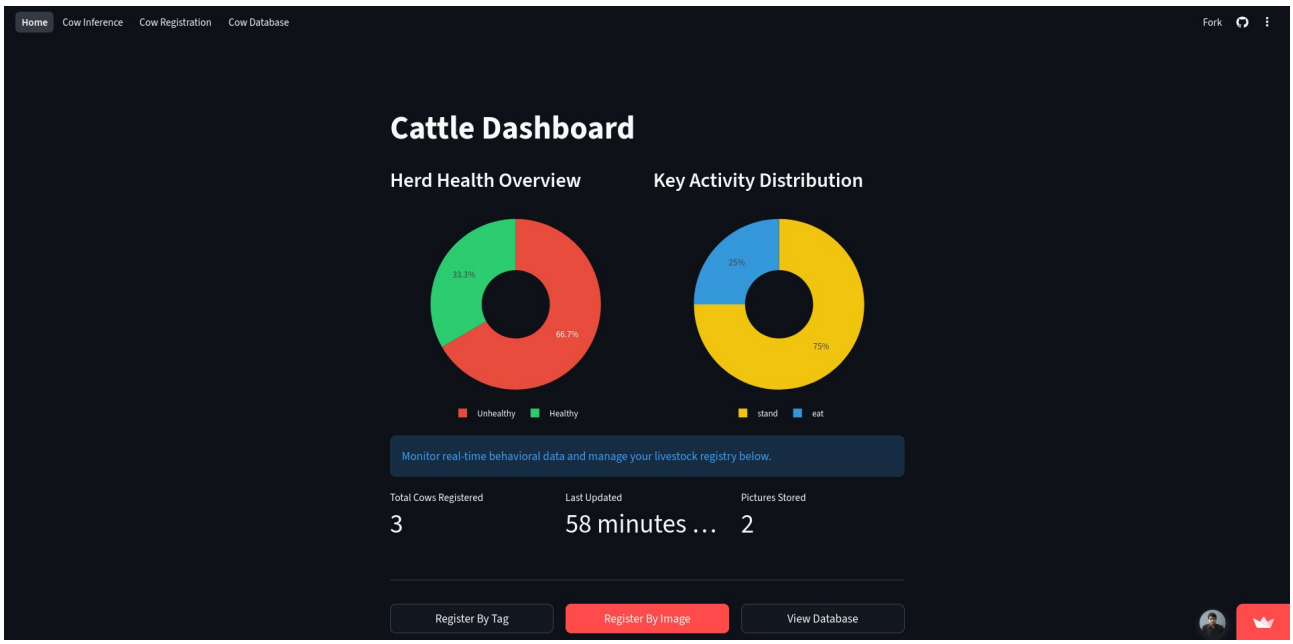
We made use of Supabase for the database, as well as their buckets for storage of images. We give each image a unique name and automatically generate a url valid for an hour to show the image when needed.

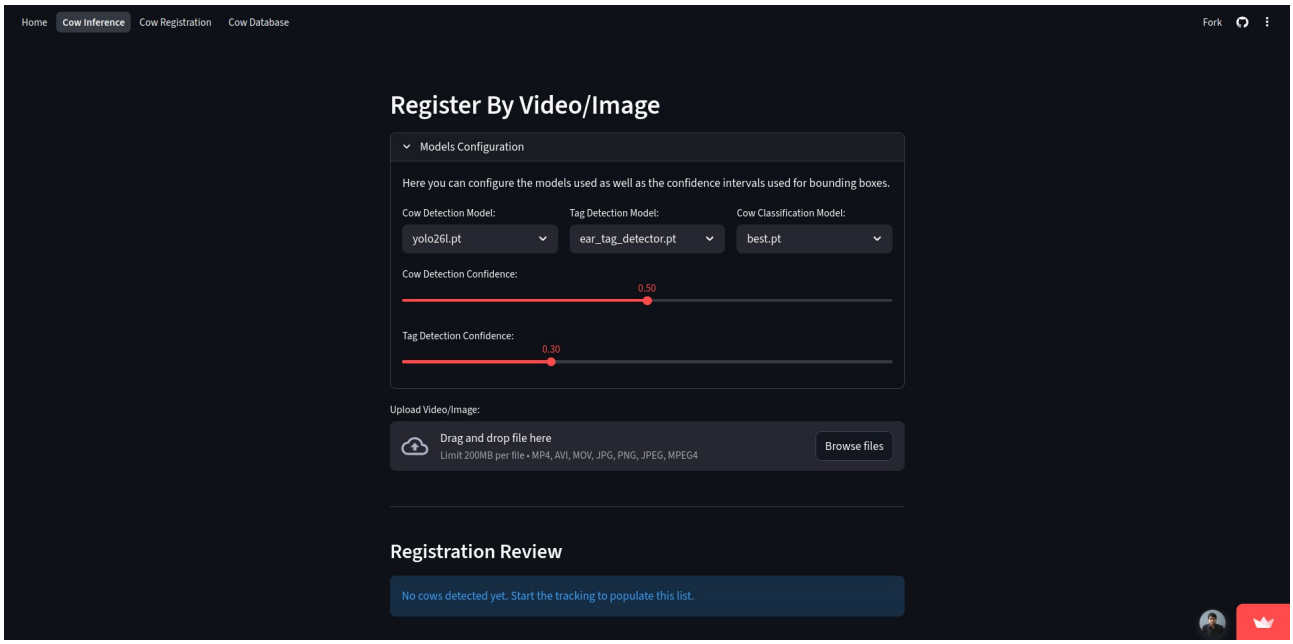
The home page shows overall cattle health as well as distribution of activity. It shows how many cattle are registered and how many images are saved, as well as the last update time for the entire database.

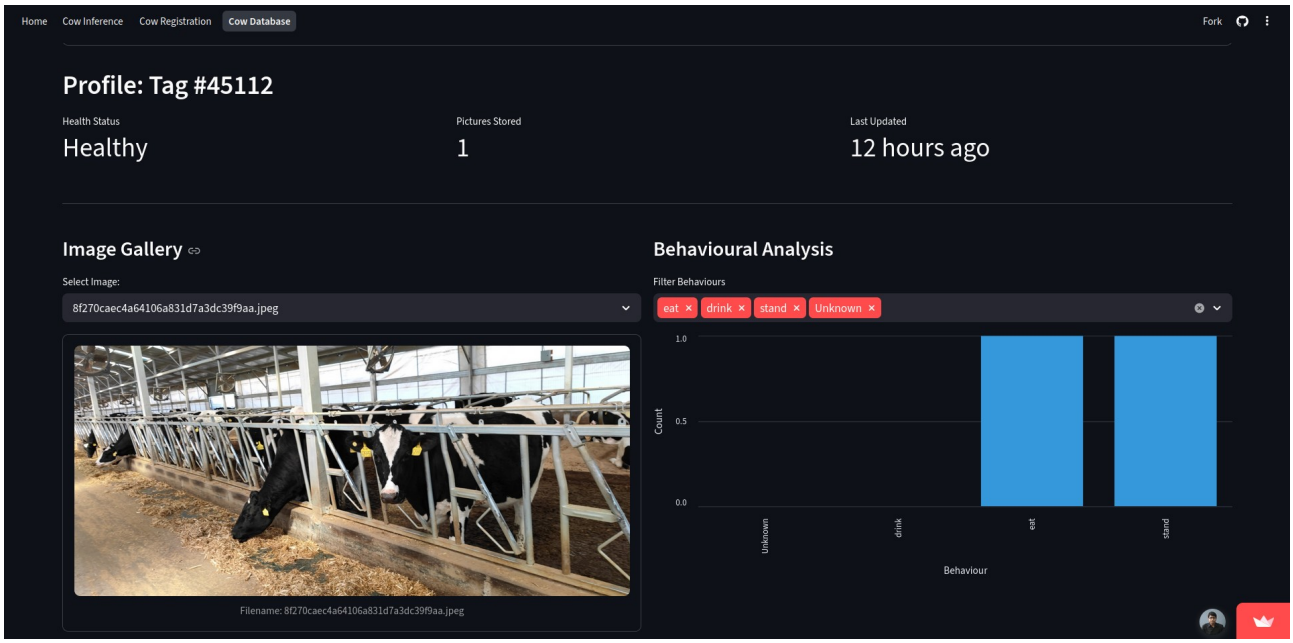
The inference page allows registration by video or image. We can choose from multiple versions of models for each type of detection or classification. By default it chooses our best models. We can also tune the confidence interval if needed. When we upload an image it shows the bounding box of the cow as well as it's ear tag id. We can edit these if needed if the model gives inaccurate results, and then save the results to the database. If the cow was already registered it simply updates it's data. Note that the image is also stored.

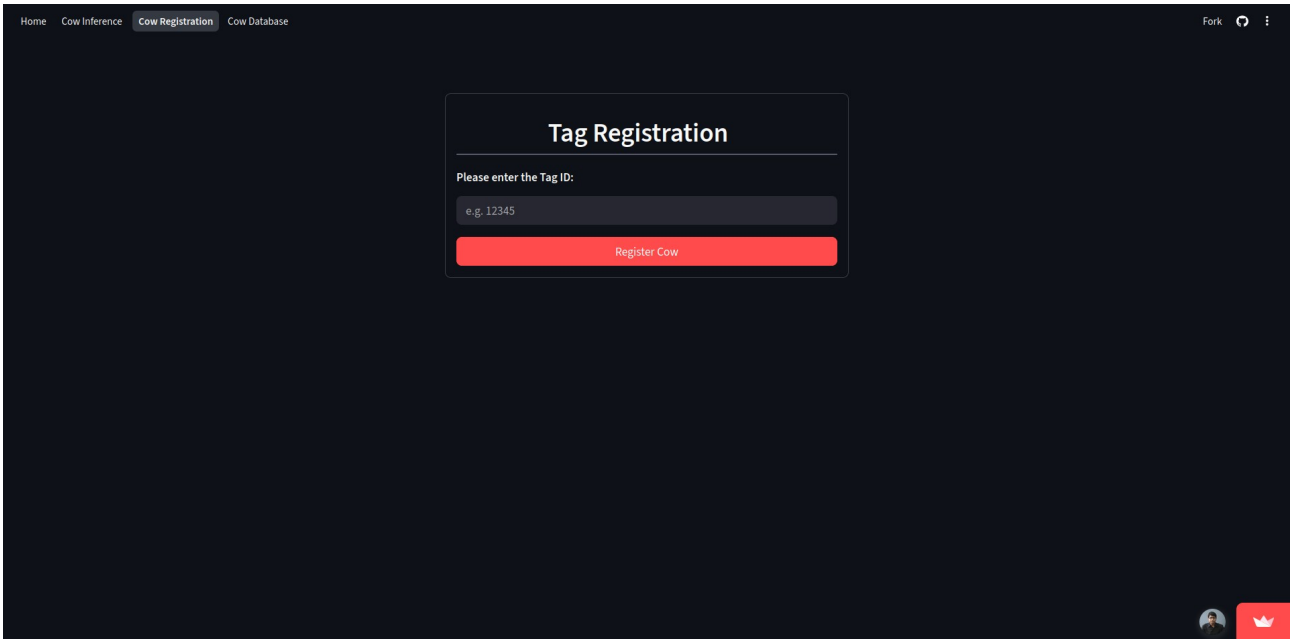
The tag registration page allows us to simply register a cow by it's tag. Simple validation is performed.

The cow database page allows viewing the raw data of the database as well as searching by tag id. When a cow is selected it shows the health of the cow, how many pictures it appears in, and when it was last updated. We can also view any of the images the cow appears in as well as view a plot of it's behaviours.


Home
Cow Inference
Cow Registration
Cow Database
Fork
Cattle Dashboard
Herd Health Overview
Key Activity Distribution
33.3%
66.7%
25%
75%
Unhealthy
Healthy
stand
eat
Monitor real-time behavioral data and manage your livestock registry below.
Total Cows Registered
3
Last Updated
58 minutes ...
Pictures Stored
2
Register By Tag
Register By Image
View Database


Home
Cow Inference
Cow Registration
Cow Database
Fork
Register By Video/Image
Models Configuration
Here you can configure the models used as well as the confidence intervals used for bounding boxes.
Cow Detection Model:
yolo26l.pt
Tag Detection Model:
ear_tag_detector.pt
Cow Classification Model:
best.pt
Cow Detection Confidence:
0.50
Tag Detection Confidence:
0.30
Upload Video/Image:
Drag and drop file here
Limit 200MB per file • MP4, AVI, MOV, JPG, PNG, JPEG, MPEG4
Browse files
Registration Review
No cows detected yet. Start the tracking to populate this list.


Home
Cow Inference
Cow Registration
Cow Database
Fork
Profile: Tag #45112
Health Status
Healthy
Pictures Stored
1
Last Updated
12 hours ago
Image Gallery
Select Image:
8f270caec4a64106a831d7a3dc39f9aa.jpeg
Filename: 8f270caec4a64106a831d7a3dc39f9aa.jpeg
Behavioural Analysis
Filter Behaviours
eat
drink
stand
Unknown
1.0
0.5
0.0
Count
Unknown
drink
eat
stand
Behaviour



Improvements Left

One major improvement left is attempting to generalize the AI model. The model currently works well for its own dataset as well as the dataset used to populate our prototype user interface. However it still struggles on random images of drinking cattle.

To this end we can change the dataset and employ two models at the same time, which could be trained on two views, top and side view, as described in one of the papers. We can also extract just a small area of the images, like the mouth area for better accuracy. We could also manually attempt to use pose estimation, as that was also mentioned in the literature review with good accuracy. However regardless the best option is to create our own dataset that is well suited for our own purposes.

For the user interface, all metrics and plots that we have made have been made at our own discretion. A farmer who would want to use this project would be more knowledgeable on what metrics would be essential for them. Thus we would want to interview a farmer for the things he uses to figure out cattle health. Additionally, we do not perform user authentication, for a proper application, this is essential. Security is also something to consider. While we have attempted to make sure that the app isn't vulnerable, as we aren't cybersecurity engineers we can only do so much. In terms of scalability as well the app should work well, but we can't be sure until we have that demand.

Furthermore, the registration methods are quite slow to use for bulk images. Inspiration should probably be taken from Roboflow for how to create a good UI for bulk registration.

Finally, note that our dataset with ear tags barely had any images of drinking cows, as well as no videos at all. Additionally, OCR is a slow technology, and it would be difficult to use with videos on any interval greater than say one frame extracted every 30 seconds.

As we only had two days to make the interface, we used streamlit but found it had a 1 GB memory limit. This meant that easyocr was extremely heavy and caused the app to crash most of the time. Preferably we would use some other framework and host it on a more capable machine ourselves.

Bibliography

[1] Lucy, “Beef Cattle Behavior Data Set,” *Kaggle.com*, 2025. https://www.kaggle.com/datasets/lucyfirst/beef-cattle-behavior-data-set/data (accessed Mar. 05, 2026).

[2] K. Li, D. Fan, H. Wu, and A. Zhao, “A new dataset for video-based cow behavior recognition,” *Scientific Reports*, vol. 14, no. 1, Aug. 2024, doi: https://doi.org/10.1038/s41598-024-65953-x.

[3] A. Zia *et al.*, “CVB: A Video Dataset of Cattle Visual Behaviors,” *arXiv.org*, 2023. https://arxiv.org/abs/2305.16555 (accessed Mar. 05, 2026).

[4] L. Yuan *et al.*, “A Review of Cattle Behavior Recognition Methods Based on Computer Vision and Deep Learning,” *Smart Agricultural Technology*, p. 101867, Feb. 2026, doi: https://doi.org/10.1016/j.atech.2026.101867.

[5] Y. Zhang, X. Li, Z. Yang, S. Hu, X. Fu, and W. Shen, “Recognition and statistical method of cows rumination and eating behaviors based on Tensorflow.js,” *Information Processing in Agriculture*, vol. 11, no. 4, pp. 581–589, Nov. 2023, doi: https://doi.org/10.1016/j.inpa.2023.11.002.

[6] P. Hanpinitsak, T. Katanyukul, N. Tonmitr, C. Suntra, S. Tanusilp, and A. Phuphaphud, “Artificial intelligence-based dairy cattle behavior recognition for estrus detection via ensemble fusion of two camera views,” *PLOS One*, vol. 21, no. 1, p. e0340999, Jan. 2026, doi: https://doi.org/10.1371/journal.pone.0340999.